

Row index

0	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	a	a
1	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	b	b
2	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	c	c
3	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	d	d
4	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	e	a
5	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	f	b
6	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	g	c
7	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	h	d
8	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	i	a
9	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	j	b
10	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	k	c
11	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	l	d
12	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	m	a
13	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	n	b
14	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	o	c
15	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	p	d

$\times$ $=$

(a) Toeplitz: Maximal non-zero diagonals

0	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	a	a
4	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	b	a
1	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	c	b
5	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	d	b
8	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	e	a
12	1	2	0	0	3	4	0	0	0	0	0	0	0	0	0	f	a
9	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	g	b
13	0	0	1	2	0	0	3	4	0	0	0	0	0	0	0	h	b
2	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	i	c
6	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	j	c
3	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	k	d
7	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	l	d
10	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	m	c
14	0	0	0	0	0	0	0	0	1	2	0	0	3	4	0	n	c
11	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	o	d
15	0	0	0	0	0	0	0	0	0	0	1	2	0	0	3	p	d

$\times$ $=$

(b) Multiplexed: Minimal non-zero diagonals